FGFR2 fusion-hotspot benchmark report

Study: brca_tcga_pan_can_atlas_2018

Abstract

This report analyzes FGFR2 gene fusions from the brca_tcga_pan_can_atlas_2018 study, covering 5 fusion events. 4/4 fusions with known frame status (100.0%) were in-frame. 2/5 fusions (40.0%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test (p=0.6, not statistically significant at alpha=0.05). No literature benchmark is configured for FGFR2.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 5 analyzed fusion events, identifying 5 distinct fusion partner genes. The most frequent partner was AP1M1, observed in 1 event.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 2/4 (50.0%) of in-frame fusions retained the domain, p=0.6 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.010989.

Domain disruption

Disruption of the C-terminal autoinhibitory tail (exon 18) was tested with Fisher's exact test comparing in-frame fusions against all others; 4/4 (100.0%) of in-frame fusions disrupted the domain, p=1 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.953047.

Junction concentration

40% of 5 fusions with this gene as the 5' partner join at exon 17, against 3% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=0.0539461, corrected for testing every exon, not statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection scanned 5 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 621 aa (exon 13; label-permutation p=0.200799). This boundary is descriptive: retained/lost status follows from breakpoint position, so a clean split exists by construction and the p-value is not evidence of selection. This is -136 aa from the nearest configured domain boundary at 757 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=2) and "not_retained" (n=3). For Frame_status, retained: 2/2 (100.0%, 95% CI 34.2%-100.0%); not_retained: 2/2 (100.0%, 95% CI 34.2%-100.0%). Welch's t-test comparing Tumor_variant_count between groups gave means of unavailable vs unavailable, p=unavailable.

Composite evidence score

Composite evidence score reported: cutpoint_proximity sub-score excluded: corrected cutpoint p-value was missing or >= 0.05, cutpoint was not determinable, or mapped breakpoints were unavailable.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: FGFR2: fusion-positive-vs-negative expression comparison skipped; no sample had both established SV assay coverage and measured expression, so no fusion-negative group could be defined.

Genomic position recurrence

Genomic position recurrence produced results for this run (4 row(s) in genomic_bin_recurrence, 1 row(s) in exact_position_recurrence, 1 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: FGFR2 has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: No jointly assessed samples for FGFR2 point_mutation (S252W); skipped.

Window detection

Window detection reported: no candidate window of the configured widths (25, 50, 100, 200) met the minimum-event-count guard (min_events_per_window=4) on both sides of the split

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
AP1M1	1	0.2	0.19590413923210936	0.002088570277522351	None	0.342380227506653	confidence_certainty, domain_disruption, domain_retention, recurrence	0.1689785548402319	1
CASP7	1	0.2	0.19590413923210936	0.002088570277522351	None	0.342380227506653	confidence_certainty, domain_disruption, domain_retention, recurrence	0.1689785548402319	2
CCDC6	1	0.2	0.19590413923210936	0.002088570277522351	None	0.342380227506653	confidence_certainty, domain_disruption, domain_retention, recurrence	0.1689785548402319	3
CTNNA3	1	0.2	0.19590413923210936	0.002088570277522351	None	0.342380227506653	confidence_certainty, domain_disruption, domain_retention, recurrence	0.1689785548402319	4
ENPP2	1	0.2	0.19590413923210936	0.002088570277522351	None	0.342380227506653	confidence_certainty, domain_disruption, domain_retention, recurrence	0.1689785548402319	5

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
427	0	2	1	2	0.0	1.0	-0.0
480	0	2	2	1	0.0	0.4000000000000001	0.39794000867203755
621	0	2	3	0	0.0	0.10000000000000002	0.9999999999999999

domain_disruption tables

frame_domain_contingency_table

4	1
0	0

frame_domain_contingency_tables_by_tumor_type

stratum	table
IDC	[4, 1], [0, 0]

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
c_terminal_tail	C-terminal autoinhibitory tail (exon 18)	5	0	0	5	0.0

domain_retention tables

frame_domain_contingency_table

2	0
2	1

frame_domain_contingency_tables_by_tumor_type

stratum	table
IDC	[2, 0], [2, 1]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	5	2	1	2	0.16967509025270758

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
11	3	5	0.6

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
AP1M1	1	1	1	1
CASP7	1	1	1	1
CCDC6	1	1	1	1

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
CTNNA3	1	1	1	1
ENPP2	1	1	1	1

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
10	121483000	121484000	2	EVT-TCGA-AN-A0AL-01-2, EVT-TCGA-D8-A13Z-01-3
10	121496000	121497000	1	EVT-TCGA-AQ-A04L-01-1
10	121503000	121504000	1	EVT-TCGA-AO-A03V-01-4
10	121515000	121516000	1	EVT-TCGA-LD-A7W5-01-5

exact_position_recurrence

chromosome	position	n_events	event_ids
10	121483698	2	EVT-TCGA-AN-A0AL-01-2, EVT-TCGA-D8-A13Z-01-3

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
767	2	1	0	False

junction_concentration tables

junction_counts

role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
five_prime	17	2	5	0.4	0.03158404596472475	2.028715521763803
five_prime	9	1	5	0.2	0.09501937466595403	0.4056193683308949
five_prime	10	1	5	0.2	0.025003340459647246	0.9247472899285746
five_prime	13	1	5	0.2	0.07189504275788348	0.506722178514524

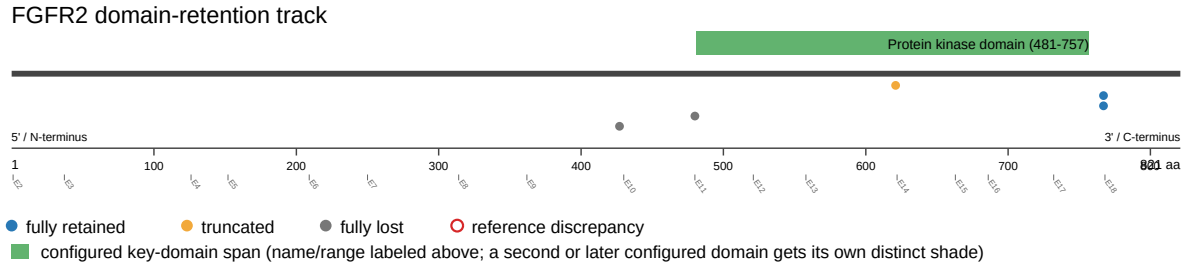
Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-AQ-A04L-01-1	TCGA-AQ-A04L-01	TCGA-AQ-A04L	Breast Cancer	IDC	FGFR2-AP1M1	AP1M1	in-frame	five_prime	121496532	13	621	False	621	True	exact match	disrupted	0.5090252707581228	True	partner_lacks_domain	PF07714					15			True			C-terminal autoinhibitory tail (exon 18)	Protein kinase domain	NA | FGFR2-AP1M1 Fusion	in-frame
EVT-TCGA-AN-A0AL-01-2	TCGA-AN-A0AL-01	TCGA-AN-A0AL	Breast Cancer	IDC	FGFR2-CASP7	CASP7	in-frame	five_prime	121483698	17	767	False	767	True	exact match	retained	1.0	False												Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		NA | FGFR2-CASP7 Fusion	in-frame
EVT-TCGA-D8-A13Z-01-3	TCGA-D8-A13Z-01	TCGA-D8-A13Z	Breast Cancer	IDC	FGFR2-CCDC6	CCDC6	in-frame	five_prime	121483698	17	767	False	767	True	exact match	retained	1.0	False												Protein kinase domain	C-terminal autoinhibitory tail (exon 18)		NA | FGFR2-CCDC6 Fusion	in-frame

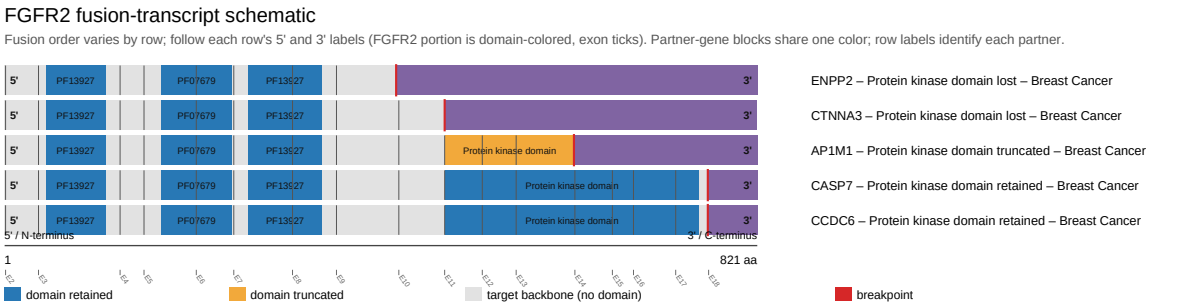
event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-AO-A03V-01-4	TCGA-AO-A03V-01	TCGA-AO-A03V	Breast Cancer	IDC	FGFR2-CTNNA3	CTNNA3	unknown	five_prime	121503790	10	480	False	479	True	hybrid junction codon: fusion-annotation reports the last complete 5-prime residue, exactly one less than this project's hybrid-junction-inclusive residue; expected convention difference, not a disagreement	lost	0.0	False																Protein kinase domain; C-terminal autoinhibitory tail (exon 18)		NA | FGFR2-CTNNA3 Fusion	frameshift
EVT-TCGA-LD-A7W5-01-5	TCGA-LD-A7W5-01	TCGA-LD-A7W5	Breast Cancer	IDC	FGFR2-ENPP2	ENPP2	in-frame	five_prime	121515123	9	427	False	427	True	exact match	lost	0.0	False														Protein kinase domain; C-terminal autoinhibitory tail (exon 18)		NA | FGFR2-ENPP2 Fusion	in-frame

Figures

domain retention outliers



fusion schematic



Partner-block colors are decorative; row labels identify partners.